

Project Report: TF-IDF Analysis of Evolution and Ethics Texts

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1 Summary

This report uses TF-IDF to compare five books that discuss the relationship between the theory of evolution and ethics. The list of books includes two scientific works written by Charles Darwin, two ethics focused works that respond to evolution by T. H. Huxley and Herbert Spencer, and one work that combines both views by Peter Kropotkin. My goal with this project was to test whether the vocabulary used within the books reflects a spectrum from science to ethics.

- **TF-IDF accurately identifies what makes each book distinctive.** Each book's top terms, ranked by TF-IDF value, match its subject very accurately, such as “males” and “females” for the topic of sexual selection in *The Descent of Man* or “village” and “guilds” for Kropotkin's history of cooperation.
- **The expected spectrum doesn't appear when comparing TF-IDF.** Apart from Darwin's two books, which had a similarity of 0.46, every other pair of books scored between 0.19 and 0.26. Words like “nature”, “life”, and “struggle”, which were commonly used in all five books, received IDF values of zero or below, which meant they didn't count toward similarity.
- **Scoring similarity by term frequency does show the spectrum.** Without IDF, the spectrum does show up. The two texts on ethics have a score of 0.52, *Mutual Aid* is about halfway between them and Darwin's books, and the similarity score appears to decrease steadily from Kropotkin to Huxley to Spencer.
- **Text cleaning difficulties.** There were many problems in the original texts, such as double hyphens being used as dashes and page citations which produced muddled and confusing data, but didn't cause any actual errors. These were only found by inspecting the outputs of each cleaning step.
- **Recommendations.** I found that TF-IDF is better suited to distinguishing documents, but does not do well at grouping related documents, especially with such a small dataset that focuses on one subject. Instead, using unweighted term frequency in this case does a better job at finding similarities between books. Including more varied books in the dataset would also make IDF more effective at grouping them.

2 Introduction

In 1859, Charles Darwin wrote *On the Origin of Species*, which proposed the idea of evolution through natural selection. This sparked a large, ongoing debate over ethics, and whether or not

qualities like morality fit into that model. This project analyzes five central texts to this debate, which I obtained for free as plaintext editions from Project Gutenberg [1] (Table 1).

ID	Title	Author	Year	Role in comparison
1228	On the Origin of Species	Charles Darwin	1859	Pure science & the baseline of the debate
2300	The Descent of Man	Charles Darwin	1871	The origin of morality & extending evolutionary science to humans
2940	Evolution and Ethics	T. H. Huxley	1893	Ethics that argues against deriving morality from evolution
46129	The Data of Ethics	Herbert Spencer	1879	Ethics built on evolution
4341	Mutual Aid	Peter Kropotkin	1902	About cooperation & using biological evidence to build the ethical argument

Table 1: Documents analyzed (Project Gutenberg’s plaintext editions).

I chose these specific books because I’m familiar with them, having read them all, and because they represent many different ideas along the spectrum in this debate. Darwin’s two books are scientific, Huxley’s and Spencer’s focus on ethics while taking opposite positions, and Kropotkin’s lies somewhere in between.

The goal of this project is to measure how these books relate to one another using TF-IDF. TF-IDF’s purpose is to turn the text in the books into vectors that can be compared. More specifically, this project asks:

- Which terms are most distinctive in each book, and do they reflect the subject discussed?
- How similar is each book to every other book, and can the differences between science and ethics be seen in that similarity?

Section 3 describes how each of the texts was cleaned, as well as how the steps of TF-IDF and cosine similarity were computed. Section 4 uses data extracted during analysis to discuss the most frequent and the most distinctive terms in each book. Beyond that, the similarity between each book is also discussed, and there’s a section about how the books were compared using only term frequency when initial results were not what was expected. Section 5 goes on to summarize my findings throughout the project, as well as talking about some potential extensions for this project that could yield more useful information. Finally, Section 6 contains my personal reflections on the whole process, and what I’d do differently next time.

3 Methodology

3.1 Data Collection

The five books were obtained from Project Gutenberg [1] as UTF-8 plaintext files. The download script uses each book’s Gutenberg ID to fetch each file from its URL (<https://www.gutenberg.org/ebooks/<id>.txt.utf-8>) and saves to the `data/raw` folder in the project. The files are downloaded once and kept locally rather than being fetched on every run.

The processing pipeline is divided into stages: trimming, normalization, tokenization, term frequency (TF), inverse document frequency (IDF), and TF-IDF. Each stage uses the output of the previous stage and writes its results to its own output folder. By storing the results of each stage on disk, I was able to inspect the data between steps, which was very helpful when debugging cleaning errors. For example, I was able to find the hyphenation error described in Section 3.3 by comparing the trimmed and normalized versions of a book.

3.2 Text Cleaning and Trimming

Some parts of the files obtained from Gutenberg are not part of the author’s text, such as a license header and footer, title pages, a table of contents, and an index. If included in the analysis, that content would add words that say nothing about the books’ content, so they were all trimmed to only be the portion of text between a start and end phrase, which I found manually for each book. In addition, the end phrase is only searched for after the start phrase is found in the text, which prevents the text trimming from cutting off early, since section titles such as the end phrase can also appear earlier, in the table of contents.

Within the books, material that was written by the authors themselves was kept, such as introductions, prefaces, and footnotes, because they are part of the main content of the book. I considered removing chapter heading lines, but they make up such a small fraction of the book’s words that they wouldn’t meaningfully affect the results.

The book that required the most extensive trimming was Huxley’s work. The Gutenberg edition is actually a collection of 5 essays, and only the first two (the Prolegomena and the “Evolution and Ethics” lecture, with its notes) talk about the relationship between evolution and ethics. The other 3 essays talk about unrelated topics like labor economics and the Salvation Army, which, if included, would have vocabulary unrelated to my comparison between evolution and ethics. While trimming, I also found a transcriber’s note saying that the third section, “Science and Morals,” is actually a reprint of Asa Gray’s 1860 essay on Darwin, not Huxley’s. After trimming, Huxley’s text is about 163,000 characters, compared to the other books which are much longer (roughly 560,000 to 1,630,000 characters).

3.3 Normalization and Tokenization

Each trimmed book was normalized line by line. The first step in normalization was casefolding, so that, for example, “Species” and “species” count as the same word. Each line was also converted to Unicode Normalization Form C so accented characters, which appeared in some quoted names and foreign phrases in the texts, always get encoded the same way. After that, all punctuation was removed with a regular expression that deletes every character that’s not a letter, digit, or whitespace.

However, deleting punctuation outright caused an issue that I only detected when running analysis. The highest IDF values in the dataset included words that don’t exist, such as “strugglehas”. By comparing the trimmed and normalized versions of *Mutual Aid*, I saw that the older transcriptions used a double hyphen instead of a dash (“struggle--has”), so deleting hyphens fused the words together. These double hyphens appeared in *Mutual Aid*, *The Data of Ethics*, and in Huxley’s text, but never in either of Darwin’s books. I fixed this bug by adding a step to replace them with a space before any of the remaining punctuation is removed.

Single hyphens were still affected by the general punctuation rule, which deletes them and joins the two halves into one token, because they’re normally used in these books to create compound terms. In *On the Origin of Species*, this included words like “fresh-water”, “parent-species”, and

“sub-species”. Treating “parentspecies” as a single term does better at preserving its intended meaning, which would be lost if it were split into “parent” and “species”. One trade-off of this method is that a compound word written with a hyphen and the same phrase written as two words are counted as two different words during analysis.

Each normalized book was then tokenized by splitting the text on whitespace. Three filters were then applied:

- **Stopword removal.** Common function words such as “the”, “of”, and “which” were removed using scikit-learn’s English stopwords list [2]. These words appear in every book, but don’t help with differentiating what the subject is. Without removal, they dominated every book’s term frequencies.
- **Single-character tokens.** *The Descent of Man* cites page numbers such as “p. 45” more than 1,000 times. Once punctuation was removed during the normalization stage, the leftover “p” tokens became the book’s most frequent terms, so I also removed all one-character tokens from the set of tokens.
- **Numeric tokens.** Page numbers, section numbers, and dates don’t carry any meaning on their own, so any tokens that only consisted of digits were also removed.

Stopword removal had the largest effect on the data. It removed about 55% of all tokens, while only shrinking the vocabulary of the texts by about 300 words.

Finally, each book’s tokens were stored as a table with one row per token and a column recording which book it came from. Then, the five books’ tables were combined into the single table used for the rest of the analysis. Even though the tables were combined, each row is labeled with its book, so term counts can be computed for each book, as well as statistics like the number of books that contain each term. Table 2 summarizes the final result.

Book	Tokens	Unique terms
On the Origin of Species	63,344	7,253
The Descent of Man	120,753	14,335
Evolution and Ethics	11,822	4,426
The Data of Ethics	40,136	6,679
Mutual Aid	42,422	9,586
All books	278,477	24,828

Table 2: Tokens and vocabulary size per book after cleaning.

3.4 Vectorization and TF-IDF

Term frequency, inverse document frequency, and TF-IDF were computed with pandas and numpy following the definitions in the project handout:

$$\text{tf}(t, d) = \frac{f(t, d)}{\sum_{t' \in d} f(t', d)} \quad (1)$$

$$\text{idf}(t, D) = \log\left(\frac{N_D}{1 + n_t}\right) \quad (2)$$

$$\text{tfidf}(t, d, D) = \text{tf}(t, d) \cdot \text{idf}(t, D) \quad (3)$$

Each book’s term frequency was computed separately. Counting how many times each unique term appears in a book gives $f(t, d)$, and dividing every term count by the book’s total number of tokens gives $\text{tf}(t, d)$. This then turns each book’s table of individual tokens into a smaller table, with one row per unique term. Term frequency will have one value for every combination of term and book, and a given term will usually have a different term frequency for each book it appears in.

By contrast, inverse document frequency has a single value per term across the whole collection of books. Because each book’s term frequency table lists every term in that book exactly once, n_t , which is the number of books that contain a given term, can be found by combining the 5 tables and counting how many times each term appears. With $N_D = 5$ books, the IDF of each term was then computed with the natural logarithm.

Finally, each book’s term frequency table was merged with the shared IDF table based on the term, and the two values were simply multiplied to find each term’s TF-IDF score in that book. The results of the TF, IDF, and TF-IDF steps are all saved to disk, so later analysis can reuse them without having to recompute earlier stages.

Because there are only 5 books in the dataset, n_t can only be between 1 and 5, so IDF can only have 5 possible values.

Books containing the term (n_t)	1	2	3	4	5
IDF	0.916	0.511	0.223	0.000	−0.182

Because of the $1+$ in the denominator of IDF, a term found in 4 of the books will have an IDF of exactly 0, and a term found in all 5 books will have an IDF that’s slightly negative. Those terms would also have a TF-IDF of 0 or below in every book, no matter how frequent they are. The effect of this on the results is discussed in Section 4.3.

3.5 Document Comparison

To compare the different books with each other, each book’s TF-IDF scores were treated as a vector. This meant reshaping the combined TF-IDF table from one row per term-and-book pair into a matrix, which had one row per book and one column for each term that appeared. Because not every book had all 24,828 terms, the matrix entries were just filled in with zero wherever they didn’t have a term. Each book can then be expressed as a vector of 24,828 TF-IDF scores.

The similarity between two books was measured using cosine similarity:

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad (4)$$

Cosine similarity measures the angle between 2 vectors, ignoring their lengths. A score of 1 means the 2 books weight their vocabulary in exactly the same proportions, while a score close to 0 means they share almost no distinctive vocabulary. This method of comparison suits this application because the books are quite different in length, so having a score that depends on which terms each book emphasizes rather than on how much text they contain is much better. The similarity was computed for every pair of books, which gives a 5 by 5 matrix.

4 Results

4.1 Vocabulary

After the cleaning, the 5 books contain 278,477 tokens in total and 24,828 distinct terms (Table 2). The size of a given book’s vocabulary mostly correlates with length, but not entirely. *The Data of*

Ethics and *Mutual Aid* are almost the same length (about 40,000 and 42,000 tokens, respectively), but *Mutual Aid* uses 9,586 distinct terms, while Spencer’s work only uses 6,679. Spencer develops an abstract argument that uses a small, repeated set of concepts, while Kropotkin uses many concrete examples relating to animal behavior, history, and specific places, all of which have their own separate vocabulary for the most part.

The most frequent terms can already be seen to reflect each book’s general subject. Darwin’s two books mostly contain biological vocabulary (“species”, “varieties”, “selection”, “males”, “females”), Spencer’s work mostly contains the vocabulary of ethics (“conduct”, “happiness”, “pleasures”), and Kropotkin’s, words relating to social life (“mutual”, “village”, “common”). Without stopword removal, however, these terms were overshadowed by words like “the”, “of”, and “and”, which were the most frequent words in every book.

4.2 Term Frequency

Table 3 lists the five most frequent terms in each book.

Book	Top five terms by TF
Origin of Species	species, varieties, forms, selection, natural
Descent of Man	males, male, female, species, females
Evolution and Ethics	nature, existence, man, state, process
Data of Ethics	conduct, life, happiness, pleasures, actions
Mutual Aid	life, mutual, village, common, new

Table 3: Most frequent terms in each book after cleaning.

Sorted term frequencies, in a way, give a quick summary of each book. “Species” alone makes up 2.3% of all tokens in *On the Origin of Species*, which is more than three times the next most frequent term, and makes sense for a book written around a single question. *The Descent of Man*’s top terms are almost all about sex, which reflects the fact that most of the book is actually about sexual selection, rather than about human origins as its title suggests. Huxley’s most frequent terms (“nature”, “existence”, “process”) are more general and abstract, which matches the essay’s philosophical treatment of the “cosmic process” of evolution.

However, term frequency on its own has a clear weakness, which is the fact that several top terms are shared across books. “Species” is a top term in both Darwin books, and “life” is in both *The Data of Ethics* and *Mutual Aid*. Term frequency does a good job at showing what a book talks about most, but it fails to show what distinguishes it from the other books.

4.3 Inverse Document Frequency

The terms with the lowest IDF are the ones that are found in all 5 books, such as “species”, “forms”, “natural”, “plants”, and “life”. These are some of the central words of the debate between evolution and ethics, which is why every book uses them. As described in Section 3.4, their IDF is slightly negative, so their TF-IDF score is negative in every book. In the case of this dataset, “species”, which is the most frequent term in *On the Origin of Species*, scores below 0 there.

By contrast, terms with the highest IDF appear in only one book. This includes words like “silurian”, “fossils”, and “fossiliferous” from Darwin’s chapters on the geological record in *On the Origin of Species*. Because many terms share the maximum IDF value, their order is completely

arbitrary, so IDF on its own only separates terms into five groups. The data will only become informative when combined with term frequency, to find TF-IDF.

4.4 TF-IDF

Figure 1 shows the 10 highest TF-IDF terms in each book, and Table 4 lists the single highest.

Book	Highest TF-IDF term	Score
Origin of Species	formations	0.00127
Descent of Man	males	0.00219
Evolution and Ethics	cosmic	0.00519
Data of Ethics	altruism	0.00233
Mutual Aid	village	0.00255

Table 4: Highest TF-IDF term in each book.

Compared to raw term frequency, TF-IDF produces much more distinctive terms. Each book’s top terms show that work’s particular contribution. In *Origin*, “formations” and “hybrids” reflect Darwin’s chapters on the geological record and on hybridism. *Descent* uses “males”, “females”, “ornaments”, and “plumage”, and words like “cosmic”, “karma”, and “nirvana” show Huxley’s comparison between evolution and Buddhist and Hindu philosophy. In addition, Spencer’s work uses words like “altruism” and “egoism” to advance his central question, while Kropotkin mentions “medieval”, “guilds”, and “communal” to talk about the history of cooperation. By using TF-IDF, the general terms that were common among every book, such as “species” and “life”, have fully disappeared.

Huxley’s top score of 0.00519 is about twice as high as any other book’s. This is likely because Huxley’s essay is short and focused on a single theme, the ‘cosmic process’, while the longer books cover many topics, which spreads their tokens across more terms.

These rankings also allow us to see remaining artifacts from the source texts, such as “fig” from figure references in *Descent*, “shewn” (Darwin’s spelling of “shown”), the names of cited authors such as “gärtner” or “rhys davids”, and “breedingseason”, which was produced by the hyphen-joining rule that’s described in Section 3.3. None of these artifacts actually change the overall picture, but they do show what the limits of the cleaning process are.

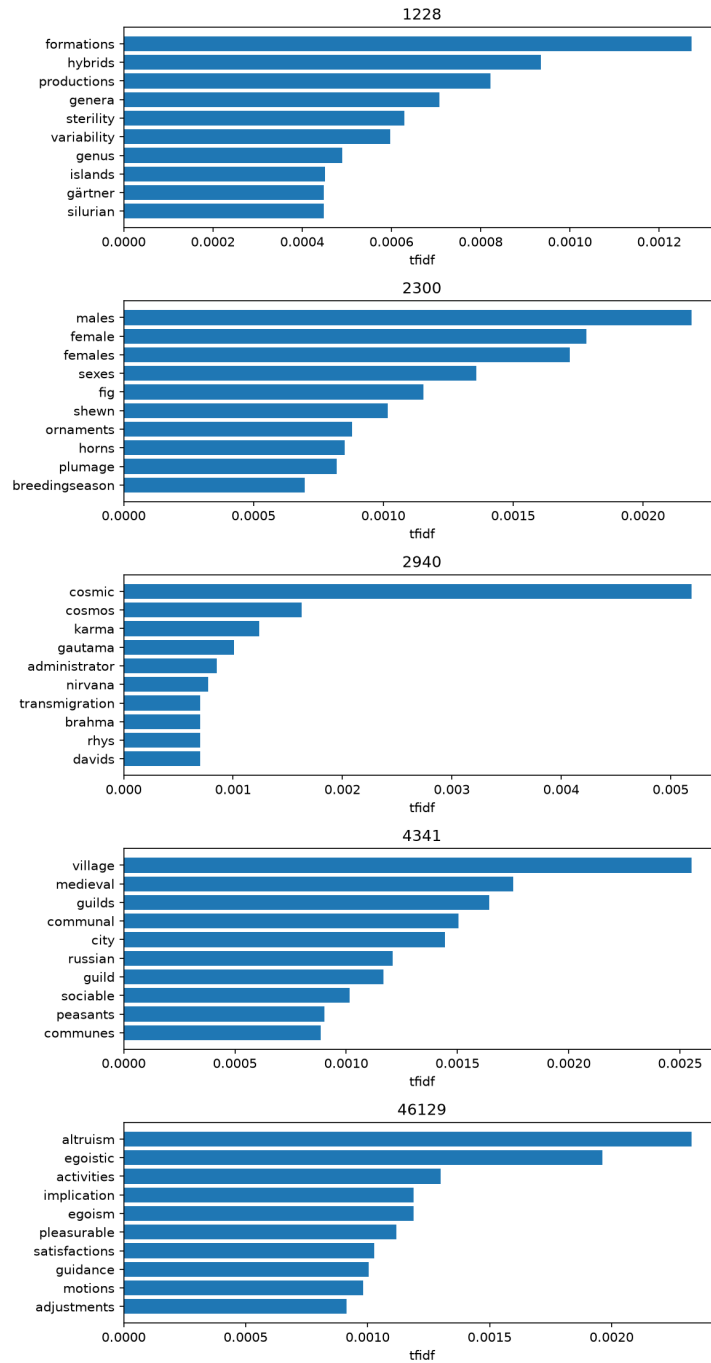


Figure 1: Top ten TF-IDF terms for each book.

4.5 Document Similarity

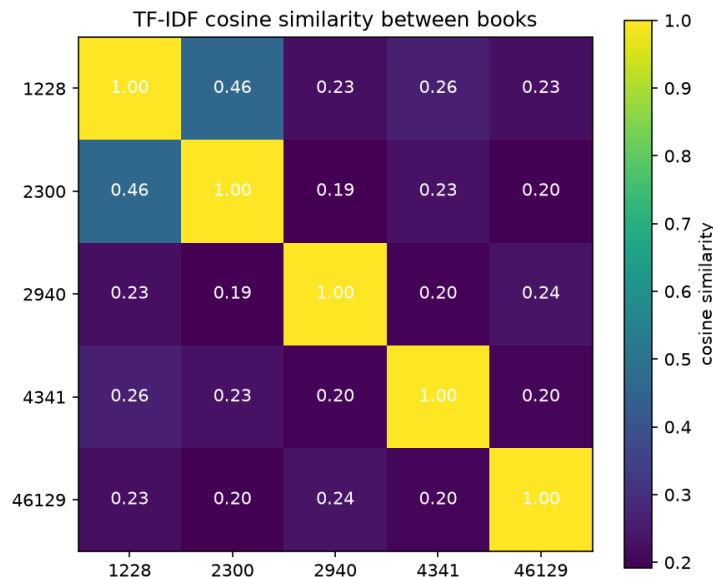


Figure 2: Cosine similarity between the books’ TF-IDF vectors.

Figure 2 shows the cosine similarity between every pair of books. The two Darwin books are the most similar pair by far, with a score of 0.46. Every other pair in the heatmap falls between 0.19 and 0.26.

The expected spectrum between science and ethics doesn’t seem to appear. The two texts that fall into the ethics genre, Huxley’s and Spencer’s, are only a bit more similar to each other (0.24) than to Darwin’s *Origin* (0.23 for both). *Mutual Aid* also doesn’t fall between the scientific and ethical texts, and its closest match is *Origin* (0.26), which is consistent with the opening chapters of the book being about cooperation among animals.

I think the most likely explanation for this lies in the IDF values. The vocabulary that these books share, words like “nature”, “life”, “species”, and “struggle”, is exactly the vocabulary that IDF scores at 0 or below, so it contributes almost nothing when computing the TF-IDF vectors. The only words that remain after that are the books’ distinctive vocabularies, and each book is distinctive in its own direction, with each one being mostly about either geology, sexual selection, Buddhism, egoism, or medieval guilds. TF-IDF is designed to separate documents, and with only five documents that share a common subject, it just separates them about equally. As a result of this, only the 2 books by Darwin, which are by the same author and are about very closely related subjects, share enough distinctive vocabulary to stand out.

4.6 Exploration

Section 4.5 suggested that similarity scores between the books were low because IDF removes the vocabulary they have in common. To test that, I also computed cosine similarity using term frequency alone, without IDF weighting. In this version, shared terms such as “nature”, “life”, and “struggle” are able to count towards similarity instead of being cancelled out.

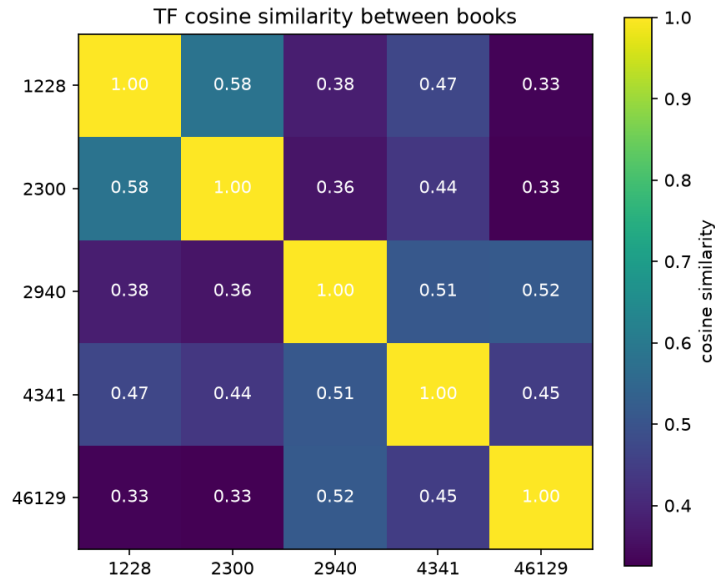


Figure 3: Cosine similarity between the books’ term frequency vectors, without IDF weighting.

As expected, every similarity score is higher (Figure 3), but the pattern shows the more important change. The two ethics texts, which before were barely more similar to each other than to Darwin under TF-IDF, now show a strong similarity. Huxley and Spencer score 0.516, and each is the other’s closest match. *Mutual Aid* falls between the two groups, scoring similarly to all 4 other books (0.436 to 0.510). It’s closer to the Darwin books than either ethics text is, but also closer to the ethics texts than either Darwin book is. And as before, Darwin’s 2 works are still the most similar pair (0.583).

The scores also form a gradient now. Similarity to *On the Origin of Species* falls steadily from *Mutual Aid* (0.469) to Huxley (0.383) to Spencer (0.332), and this order is also seen for *The Descent of Man*. This matches the spectrum these books were chosen to represent, with Darwin’s focus on science, Kropotkin’s biologically grounded argument for cooperation, Huxley’s ethics written in direct response to evolution, and Spencer’s more abstract ethical system.

This confirms that the spectrum between science and ethics is present in the texts, but it’s more present in their shared vocabulary rather than in their distinctive vocabulary. The TF-IDF method shows what makes each book unique from the others, which is why it separated the books almost equally, while term frequency shows how the books use a common set of vocabulary, which is where their relationship to each other appears. In general, deciding which method is appropriate depends on whether the goal is to distinguish the documents or to group them together.

5 Conclusion

This project used TF-IDF to examine five books that are connected by the question of how the theory of evolution relates to ethics. Two of the books are Darwin’s scientific works, two are ethical works that respond to evolution, and *Mutual Aid* combines biological evidence with an ethical argument. The goal of this project was to see whether the vocabulary of these books reflects the spectrum between science and ethics.

TF-IDF identified the distinctive content of each book very accurately. Each book’s highest-

scoring terms correspond to its actual subject. These are the geological record and hybridism in *On the Origin of Species*, sexual selection in *The Descent of Man*, Huxley’s comparison of evolution with Buddhist and Hindu philosophy, Spencer’s opposition of egoism and altruism, and Kropotkin’s history of cooperation in medieval guilds and villages. The fact that these results match each book’s known subject is the main evidence that the cleaning and computation were done correctly.

However, TF-IDF didn’t reveal the spectrum between science and ethics that was expected. Except for Darwin’s two books, every pair of books was about as dissimilar as every other pair of books. This is because words like “nature”, “life”, and “struggle” are given an IDF value of zero or below because they appear in all five books. Repeating the same comparison with just term frequency also confirmed this relationship, as without IDF the ethics texts formed a pair, *Mutual Aid* fell between them and Darwin’s books, and the books formed a gradient in similarity from science to ethics. This finding shows how the relationship between the books is more in how they use a common vocabulary, rather than in distinctive words they use. TF-IDF is very well suited for telling documents apart, but, especially with a small dataset, it also hides the connections that group them together.

Several extensions could also strengthen these conclusions:

- **More documents.** With only 5 books, IDF can only take 5 values, and any term found in 4 or 5 books will be effectively discarded. A larger collection of documents, including books outside the debate of science and ethics, would help IDF separate more common English vocabulary from vocabulary that’s only commonly used within the debate.
- **Bigrams.** Phrases like “natural selection” and “mutual aid” get each book’s ideas across much more accurately than separating them into individual words, which could be shared by other books.
- **Stemming.** *The Descent of Man*’s top terms included “males”, “male”, “females”, and “female” as four separate terms. Stemming would combine similar forms like these and give a clearer picture of what each book emphasizes.
- **Chapter-level analysis.** Treating chapters as documents could also show how arguments develop throughout a book, such as *Mutual Aid*’s progression from discussing animal behavior to human history.

6 Reflection

6.1 Something I Understand Differently Now

When I started this project, not having much experience with implementing algorithms like TF-IDF, I expected that to be the hardest part. I didn’t expect the data cleaning to take such a long time. In reality, the TF-IDF functions were actually very quick to implement and worked without much debugging. Most of my time spent programming was on fixing problems with the text, such as stopwords being the top terms in each book before I implemented stopword removal. Even after that, I still ran into the issue of “p” being the most frequent term in *The Descent of Man*, which was leftover from more than a thousand page citations. Another issue that I ran into after that was words being fused together, like “strugglehas”, which polluted the list of highest-IDF terms instead of showing the actual rarest words.

None of these bugs caused errors when running the code, but they could be seen in the analysis results that looked wrong. When I initially saw these results, I hadn’t considered that the intermediate steps could cause such large differences in the final output, so seeing these types of bugs

made me think of cleaning as a part of the analysis that's just as important as the computation part.

6.2 A Decision I Made

After finding that double hyphens were causing words to be fused together, I chose to replace them with spaces. However, I realized that it probably wouldn't be as simple of a fix for single hyphens, because they're usually used in writing to make new compound terms, rather than being used as grammatical markers like double hyphens are. If I replaced them with spaces like I did for double hyphens, it would split those compound words up and lose a lot of the intended meaning behind them. Meanwhile, deleting them would join the two parts of the compound word into a single one, which could also be harder to read.

I realized that single hyphens were much more common in all five books, while double hyphens never appeared in either Darwin book, which suggested they were being used differently. In *On the Origin of Species*, words like "fresh-water", "parent-species", "sub-species", and "rock-pigeon" all seem like compound terms that have meaning behind them that's more than the two words on their own. Because of that, I chose to just join the words together and make words like "parentspecies", which has a specific meaning that would be lost if it were split up into "parent" and "species".

During analysis, "breedingseason" was listed as one of the top TF-IDF terms in *The Descent of Man*, which could cause some confusion. I think I'd still choose to keep the words together if I were to do it again, because I think it's important to maintain the author's intended meaning when running this statistical analysis on the texts. I would probably find a smarter way to merge them together, because "parentspecies" without a hyphen looks a little ugly, but overall I think it's a better choice than splitting the words up.

6.3 A Moment of Uncertainty

When running my analysis initially, I expected the similarity scores to show a spectrum between Darwin's books and the ethics books. Instead, every pair of books apart from Darwin's, which scored 0.46, scored between 0.19 and 0.26. I wasn't really sure if that was a bug or not, because the words my analysis script listed as having the highest and lowest TF-IDF seemed like they matched the books subject.

By looking at the IDF values, I saw that words that the books shared, which included "nature", "life", and "struggle", all had IDF values below 0, which meant they contributed almost nothing to the final comparison. I ran the same comparison using only term frequency to test that explanation, and the gradient I expected showed up. Through my exploration, I learned that the lack of an actual gradient in the TF-IDF similarity graph was actually an important finding that helped me explore the data even more.

6.4 Looking Back

If I were starting this project again, I would probably choose more books with more varied topics. With only five books, IDF could only take five different values, so any word that appeared in almost all of the books had IDF values of zero or below. And because all of the books in the dataset are about the same general topic, those words they shared were almost always related to that debate.

Including more books in the dataset would help IDF distinguish commonly used English words from words that are only commonly used within that debate. A TF-IDF similarity heatmap might also show more of a spectrum with a larger dataset because of this reason.

References

- [1] Project Gutenberg. <https://www.gutenberg.org>. Plaintext editions of ebooks 1228, 2300, 2940, 4341, and 46129.
- [2] scikit-learn developers. scikit-learn: ENGLISH_STOP_WORDS. <https://scikit-learn.org>. English stopword list used for stopword removal.